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A Review of Model Predictive Direct Speed Control for PMSMs: Classification, Robustness and Comparative Analysis

Hao Fan, Tianzhe Yang, Baoyang Liu, Chaochen Gu, Xinping Guan

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ABSTRACT

Permanent magnet synchronous motors (PMSMs) are widely used in high-performance servo systems and electric transportation, placing higher demands on the dynamic response, steady-state accuracy and robustness of speed control systems. Model predictive direct speed control (MPDSC), which directly optimizes the rotor speed based on a predictive model, exhibits significantly superior overall performance compared with traditional cascaded controllers. This paper presents a systematic review of MPDSC methods for PMSMs. Within a unified predictive control framework, existing research is categorized into four types: finite control s-et MPDSC (FCS-MPDSC), deadbeat-based MPDSC (DB-MPDSC), continuous control set MPDSC (CCS-MPDSC) and explicit MPDSC (EMP-DSC). Focusing on key problems, representative research works and their improvement strategies are surveyed. After that, the two common issues of weighting factors and robustness are systematically discussed. Furthermore, a comparative analysis of typical methods is conducted to clarify their respective features and application scenarios. Finally, the main challenges and development trends of MPDSC in different terms are discussed. This review aims to provide systematic references and technical guidance for further research and practical applications of MPDSC methods.

1 Introduction

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of applications, fast and accurate speed regulation remains critical for ensuring operational stability and production efficiency [1-3].

As depicted in Figure 1, these applications typically encompass precision manufacturing, automated production lines, autonomous systems and heavy-duty drives. For example, drive systems in computer numerical control (CNC) machine tools must maintain precise speed tracking under highly varying cutting forces to ensure superior accuracy and surface quality. Similarly, motors in assembly robots performing highly dynamic multi-axis synchronized movements require rapid transient responses and overshoot-free speed control for the precise alignment and handling of components.

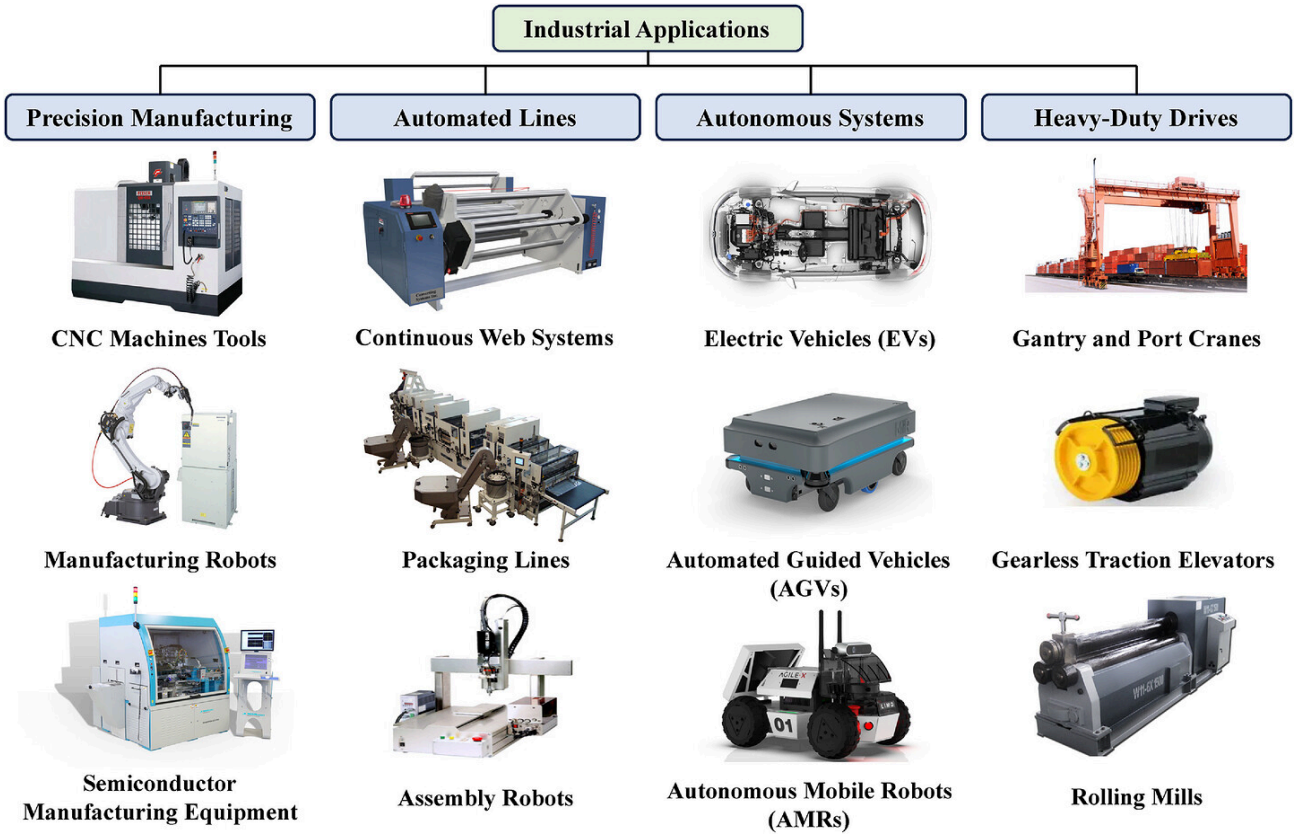


FIGURE 1

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Advanced industrial applications requiring high-dynamic PMSM speed regulation.

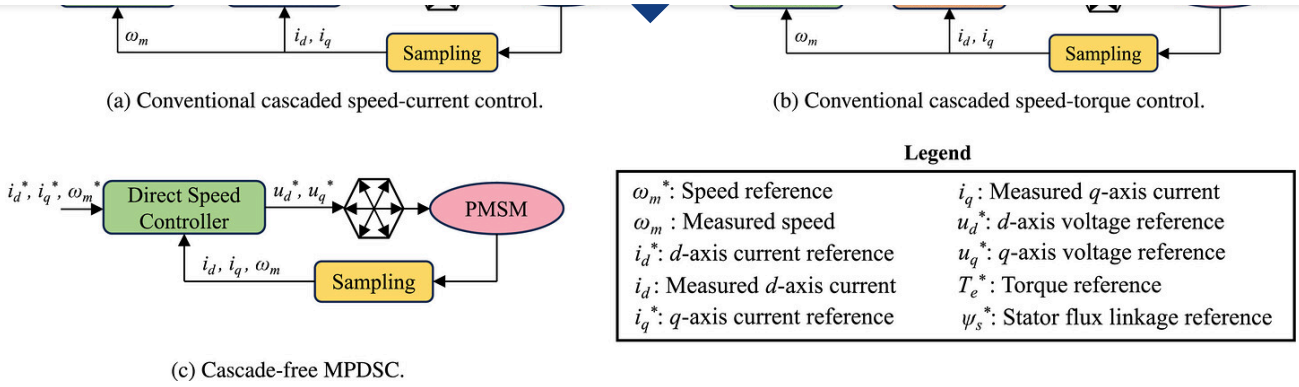
From an industrial perspective, speed controllers deployed in such scenarios must satisfy several stringent requirements. First, rapid dynamic responses and high steady-state accuracy are essential to precisely track speed references even amidst sudden load variations or high-speed operations. Second, the capacity for systematic constraint handling is crucial. Industrial drives must strictly operate within the physical limits of current, voltage and torque to prevent hardware degradation. Third, strong robustness against internal parameter mismatches and

lifespan and enhance overall energy efficiency. Low tuning complexity and a manageable real-time computational burden are also indispensable in industrial applications [4, 5].

To meet these rigorous industrial demands, various control strategies have been developed across academia and industry. Traditionally, cascaded proportional–integral (PI) controllers have served as the widely accepted industrial standard due to their structural simplicity. Such conventional frameworks typically employ an outer speed loop alongside an inner current or torque loop, as represented by field-oriented control (FOC) [6] and direct torque control (DTC) [7, 8]. Nevertheless, these conventional approaches exhibit inherent limitations in handling multivariable coupling, system nonlinearities and physical hard constraints. To address these drawbacks and better leverage system models and constraints, model predictive control (MPC) has gradually emerged over the past decade as a major research direction in motor drive systems [9]. In MPC-based drive systems, the future behaviour of the motor-inverter system is explicitly predicted using mathematical models. Subsequently, the optimal control action is selected over a finite prediction horizon according to a designed cost function, which seamlessly incorporates multiple constraints. This framework has been successfully applied to PMSM current control (MPCC), torque control (MPTC) and conventional speed control (MPSC), including both finite control set (FCS) and continuous control set (CCS) categories [10]. Regarding speed control, cascaded predictive control structures offer an appealing advanced alternative [11]. This method applies MPC to both the outer speed and inner current loops, remarkably improving dynamic responses and systematic constraint management. However, it incurs a substantial computational burden. Therefore, a widely adopted compromise involves implementing MPC in only one loop, representing an inherent trade-off between computational effort and control performance [12].

The typical cascaded control structures are illustrated in Figure 2a,b. While effective, they are fundamentally constrained by inherent inner-outer loop bandwidth limits, dictating that the speed loop must be significantly slower than the current loop. To overcome this limitation and enhance transient dynamics, model predictive direct speed control (MPDSC) stands out as a compelling single-loop alternative. Unlike cascaded designs, MPDSC eliminates the current loop as depicted in Figure 2c. It integrates speed tracking and system constraints directly into a unified cost function to evaluate and select the optimal inverter switching states or voltage vectors. By removing cascaded delays, MPDSC responds rapidly to sudden load disturbances or step speed commands, boosting the speed loop bandwidth to match that of traditional inner loops and achieving excellent dynamic responses. Furthermore, it demonstrates remarkable flexibility in multi-objective optimization. Hard physical constraints and secondary goals can be seamlessly integrated without relying on the complex anti-windup or decoupling networks required by conventional methods. Overall, the compact structure and superior transient capability of MPDSC make it an ideal choice for next-generation demanding industrial applications [13].

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(a) Conventional cascaded speed-current control.

(b) Conventional cascaded speed-torque control.

(c) Cascade-free MPDSC.

FIGURE 2

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Block diagrams of typical PMSM speed control paradigms.

Although extensive research efforts on MPDSC have been reported, to the best of the authors' knowledge, the research community still lacks a thorough review specifically dedicated to PMSM drives and focusing on direct speed control. Existing review works on MPC for motor drives predominantly focus on current or torque control [14-16]. Some reviews specifically discuss FCS methods in particular application scenarios, such as linear motors [17] and multiphase motors [18, 19]. However, these studies typically provide only a brief overview of MPDSC, and lack systematic comparisons among different strategies [13]. [20] examines MPC methods, including MPDSC, but limits its scope to weighting factor design under the FCS framework. Although [21] offers a comprehensive review of MPC strategies for AC motor drives, it does not systematically categorize or conduct an in-depth analysis of MPDSC. Moreover, the inclusion of mechanical dynamics significantly increases system dimensionality, rendering traditional analytical approaches based on unified frameworks inadequate for addressing the complexities of MPDSC. To some extent, this gap hinders researchers and engineers from fully grasping the current state of development in this field and selecting suitable MPDSC strategies for specific applications. Table 1 provides a systematic comparison between existing representative reviews and the present work.

TABLE 1. Review coverage comparison of related review papers.

Related work	Includes MPDSC	Focuses on MPDSC	Classification	PMSMs	Cost functions	Weighting factors	Robustness	Simulati
[13]	✓	✓	✓					✓
[14]				✓	✓	✓	✓	✓
[15]	✓			✓	✓	✓	✓	✓

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		MPDSC						
[16]				✓			✓	✓
[17]				Linear	✓	✓	✓	
[18]				Six-phase	✓			✓
[19]				Multi-phase	✓	✓		
[20]	✓			✓	✓	✓		
[21]	✓		✓	✓		✓	✓	

Motivated by above considerations, this paper provides a systematic overview of MPDSC methods for PMSM drive systems. The main contributions of this paper are summarized as follows.

From a unified perspective, MPDSC methods are classified into four major categories: finite control set MPDSC (FCS-MPDSC), deadbeat-based MPDSC (DB-MPDSC), continuous control set MPDSC (CCS-MPDSC), and explicit MPDSC (EMP-DSC).

Representative methods in each MPDSC category are systematically reviewed and analysed.

Common issues including weighting factors and robustness are systematically reviewed.

A comparative analysis for each MPDSC category is conducted based on simulation and detailed theoretical analysis. Practical guidelines for method selection under different application scenarios are provided.

Future research potentials are discussed, including refinement of stability theory, multi-step prediction, dimension reduction for EMP-DSC, integration with data-driven methods, and application in other machine types, with the aim of providing insights for subsequent research.

The structure of this paper is organized as Figure 3. Section 2 classifies MPDSC into four categories: FCS-MPDSC, DB-MPDSC, CCS-MPDSC and EMP-DSC. Sections 3, 3.1, 3.1.1-3.1.3, 3.2, 3.3, 4, 4.1, 4.1.1-4.1.3, 4.2, 4.2.1-4.2.3, 5 further discuss these four MPDSC categories in detail. Sections 6 and 7 explore common challenges in MPDSC, namely weighting factor tuning and elimination, as well as robust problem caused by model parameter mismatches. Section 8 presents a comparative analysis based on simulation and experiment results. Section 9 outlines future research potentials. Section 10 concludes this paper.

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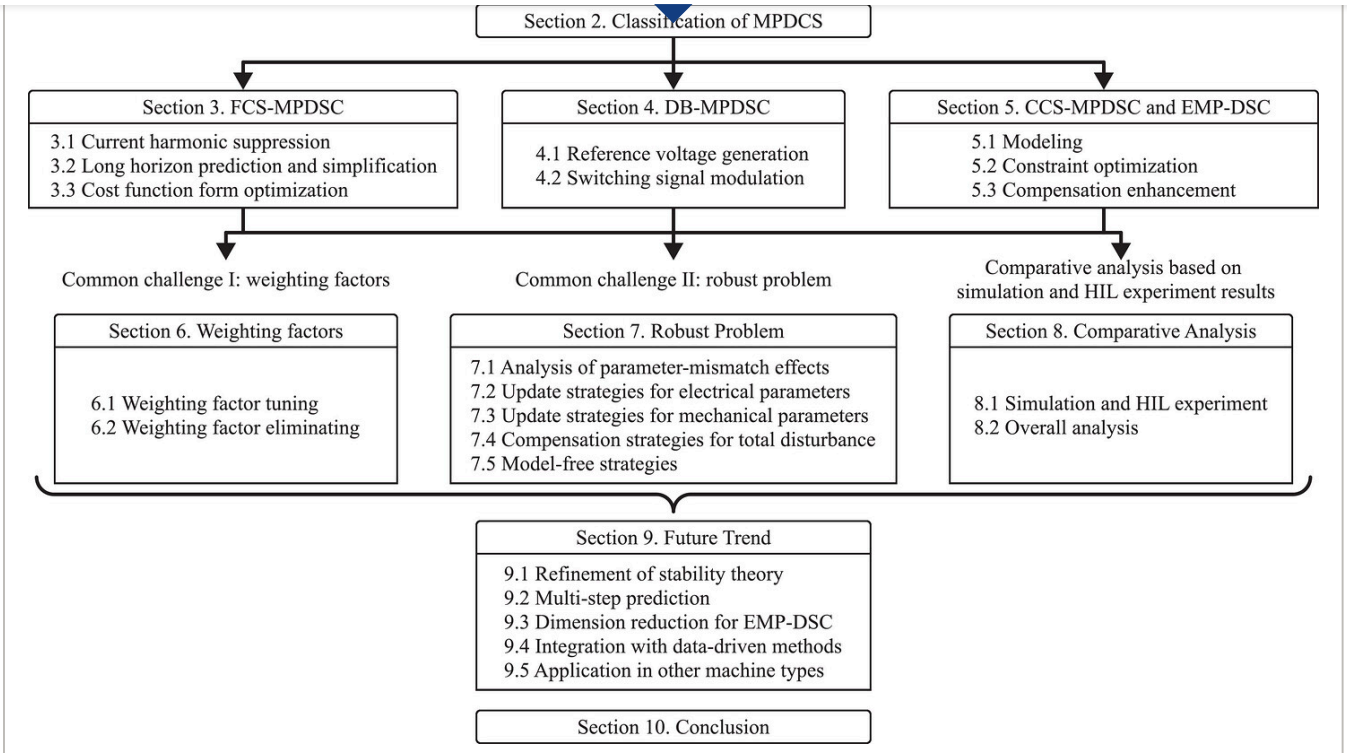


FIGURE 3

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Structure of this paper.

2 Classification of MPDSC

Among PMSM drive research, a cascaded “speed loop-current loop” structure is widely used. Specifically, the outer speed loop generates the reference current on q-axis, noted as i_q^*, based on the speed error, whereas the reference current on d-axis, noted as i_d^*, is generated by incorporating field-weakening or maximum-torque-per-ampere (MTPA) strategies. The inner current loop achieves fast tracking of current based on the reference current [22-24].

Aiming to overcome limitations of conventional cascaded structure, MPDSC has been proposed in recent years, abandoning the traditional current-centric cascaded philosophy and establishing a direct mapping from the speed reference to the voltage vector or switching state based on a unified prediction model [25-36]. Compared with MPCC, MPDSC lifts the optimization objective from the current-control layer to the speed-control layer. Using electromagnetic and mechanical equations, MPDSC performs receding horizon prediction of speed and current behaviours in the future. At each sampling instant, the controller minimizes the speed tracking error directly, and outputs the corresponding optimal control signal. With MPDSC, the relationship among speed, torque and current, is no longer realized through an explicit cascade of speed loop and current loop. Instead, it is naturally embedded in the prediction model, the cost function and the constraints. A typical cost function takes the one-step-ahead speed error as the primary term,

state ripple and switching losses.

Drawing on existing research, MPDSC methods can be broadly categorized into four categories, namely finite control set MPDSC (FCS-MPDSC), deadbeat-based MPDSC (DB-MPDSC), continuous control set MPDSC (CCS-MPDSC) and explicit MPDSC (EMP-DSC). As shown in Figure 4, these four MPDSC methods have distinct design features regarding prediction-model construction, solving strategy and implementation complexity.

FCS-MPDSC: FCS-MPDSC directly treats the inverter's finite switching states as discrete control inputs. During each sampling period, all candidate switching states are traversed, with cost functions for each candidate evaluated based on both speed and current prediction. Following that, the optimal switching state could be determined and then applied. FCS-MPDSC can achieve a fast response without introducing an independent speed loop. However, similar to FCS-MPCC, FCS-MPDSC faces challenges including current ripple and the effective switching frequency [25-27].

DB-MPDSC: Starting from a discrete-time model, DB-MPDSC aims to derive an analytical control method ensuring that the speed error tends to zero within a finite number of steps, thereby establishing a direct mapping from the speed reference to the d–q axis voltage references. Instead of solving an optimization problem online, DB-MPDSC produces a near-optimal voltage reference with the deadbeat principle, and then employs SVM or carrier-based PWM technique to determine the switching signal. DB-MPDSC exhibits advantages of smaller current harmonics than FCS-MPDSC and no need for weighting factors. However, it does not support multi-objective optimization and has relatively weak constraint-handling capabilities [28-30].

CCS-MPDSC: CCS-MPDSC formulates the speed regulation problem as a constrained finite-time optimal control problem. At each sampling instant, a continuous reference voltage vector is obtained by solving a quadratic programming (QP) problem online, which is then modulated to generate switching signals. Current harmonics of CCS-MPDSC are significantly lower than those of FCS-MPDSC. Moreover, its optimization-based framework systematically handles multi-variable constraints. However, the online solution of QP imposes a heavy computational burden, particularly in applications requiring long prediction horizons to cover mechanical dynamics. This computational demand remains the primary obstacle to its widespread application on resource-constrained embedded platforms [31-33].

EMP-DSC: As for EMP-DSC, the constrained MPDSC optimization problem is solved parametrically offline. The state space is partitioned into a set of polyhedral regions, and piecewise affine parameters are stored for each region in advance. Only limited computation is needed online to identify the state region and look up the pre-calculated parameter table, obtaining a near-optimal voltage reference. Compared with CCS-MPDSC, the explicit approach substantially reduces real-time computational burden and retains features of predictive optimization. However, EMP-DSC entails large requirement of memory, increases the system complexity with a high dimension, and challenges the robust design [34-36].

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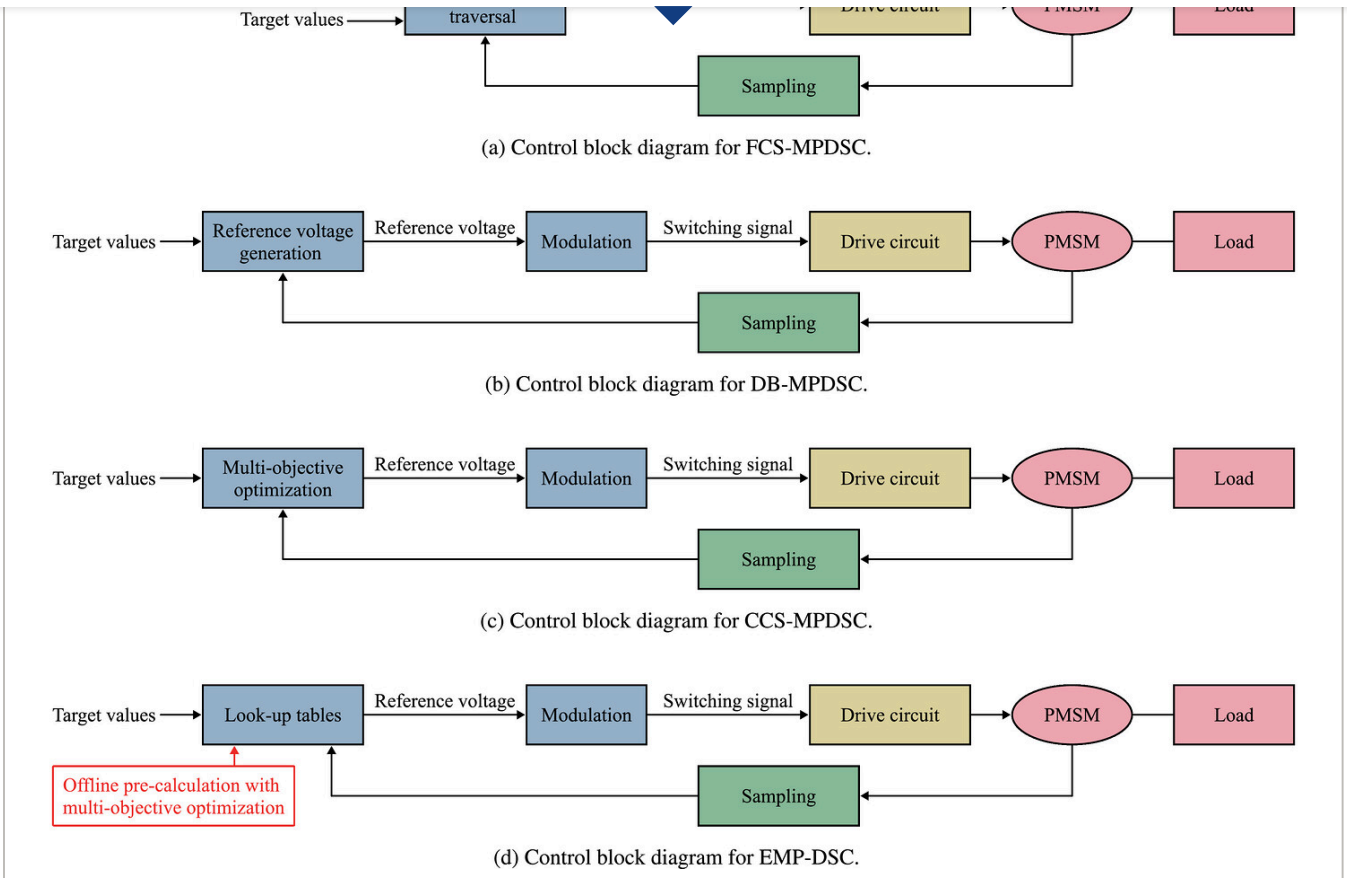


FIGURE 4

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Control block diagrams for categories of MPDSC method.

In summary, with the evolution from MPCC to MPDSC, the MPC optimization layer is lifted from the current loop to the speed loop, and researchers are attempting to coordinate the dynamic and constraints for both electrical and mechanical subsystems within a unified prediction-and-optimization framework. Sections 3, 3.1, 3.1.1-3.1.3, 3.2, 3.3, 4, 4.1, 4.1.1-4.1.3, 4.2, 4.2.1-4.2.3, 5 subsequently analyse these four categories of MPDSC methods in detail.

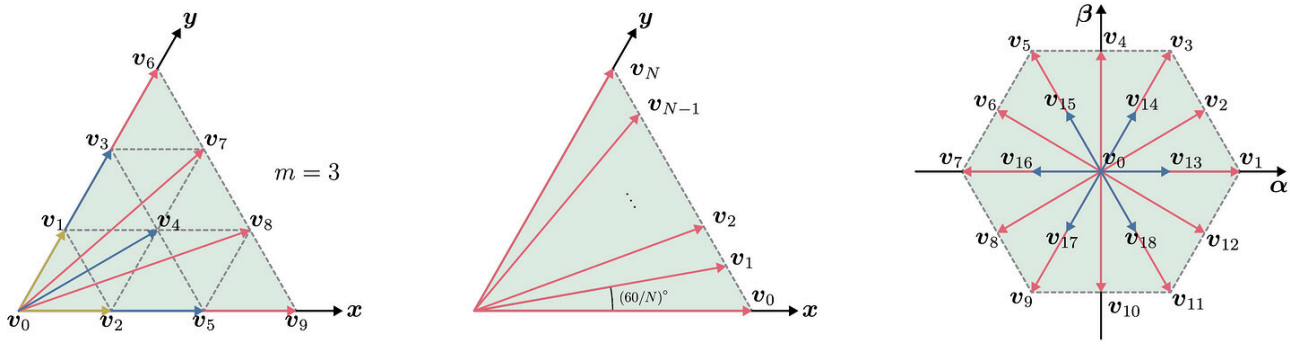
3 FCS-MPDSC

3.1 Current Harmonic Suppression

Traditional FCS-MPDSC applies only one voltage vector within a single sampling period, resulting in a stepped inverter output voltage. Consequently, the equivalent switching frequency varies with operating conditions, introducing considerable high-frequency harmonics and torque ripple in the stator currents. Investigated approaches include extended vector-set method [37, 38], optimized cost function design [39, 40] and other techniques regarding voltage smoothing [27, 41].

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The extended vector-set method aims to enlarge the candidate set of optimal vectors. The newly introduced vectors are synthesized from basic vectors, which do not physically exist as independent switching states, and named as virtual vectors as a result. For three-phase systems, virtual vectors are generally limited for the purpose of expanding the finite set. But for multiphase systems, virtual vectors can be used to suppress specific-order harmonics [42, 43]. In [44], an adjustable factor is introduced to subdivide all basic vectors evenly, and the completed virtual-vector set is obtained by pairwise linear combinations among the subdivided base vectors. And then for each virtual vector, the corresponding basic-vector pair and duty ratio can be easily determined by locating the region of the given virtual vector in the x–y plane. [45] follows a similar idea but in a reverse order, which locates the region first and constructs a virtual-vector fitting next. This method can be essentially viewed as a combination of the finite-set approach and the deadbeat method. In [37, 46], the virtual-vector set is built from six vector groups defined in a new subspace to synthesize a zero-magnitude vector. In [38], 19 vectors are synthesized using three basic vectors, eliminating weighting factors and employing a voltage-vector pre-selection strategy to improve the computational efficiency, achieving direct speed control with favourable performance. Typical construction methods for virtual-vectors are shown in Figure 5.



(a) Method used in [44]. (b) Method used in [45]. (c) Method used in [38].

FIGURE 5

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Typical construction methods for virtual-vectors.

3.1.2 Optimized Cost Function Design

Harmonic suppression can also be realized via the cost function. [39] integrates a high-frequency switching penalty term into the cost function. Some terms regarding torque and flux terms are also commonly integrated into the cost function in recent years [40, 47, 48]. In particular, [40] points out that, since the estimated electromagnetic torque is not constrained, the selected voltage vector may cause abrupt variation in torque and speed during the start-up period or the process of deceleration and acceleration. Therefore, a protective limitation term for torque could be included in the cost function. To simplify cost function design, [49]

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suppression. This transient reference improves the dynamic response of speed, suppresses current harmonics in steady state and simplifies the factor-tuning task. [50] generates q-axis current reference by combining position error proportional feedback with disturbance feedforward compensation, thereby improving current quality while enhancing system robustness. Aiming to distinguish the steady-state and transient condition based on the speed error, [51] integrates a sliding-mode term into the cost function and designs distinct cost functions for steady-state and transient conditions respectively: a speed-error term is used for steady-state condition, whereas a maximum-torque term is used for transient condition.

3.1.3 Other Methods

Alternative methods have been explored for harmonic suppression. Shown as Figure 6, [27, 41] propose a voltage-smoothing method: the candidate voltage vectors are dynamically generated as linear combinations of conventional active voltage vectors and the previously applied voltage vector. This method essentially functions as a low-pass filter for the voltage command signal, demonstrating significant harmonics suppression compared with traditional FCS-MPDSC. Based on this [52], goes further to propose that the smoothing factor should consider both transient and steady-state performance, introducing an adaptive smoothing rate, namely the smoothing factor is reduced under transient conditions to guarantee transient performance, and increased in steady-state to suppress harmonics. Additionally, switching harmonics and computational burden are reduced by restricting the candidate set at the next sampling instant. In [53], a local switching-state candidate set is constructed based on the previously selected optimal vector, avoiding explicitly inserting a switching-frequency penalty term to the cost function, and regulating the switching frequency indirectly. Similar ideas are adopted in [25, 54-56], where the switching frequency is reduced and the number of candidate vectors is also downsized from 8 to 4, with a further lower computational burden.

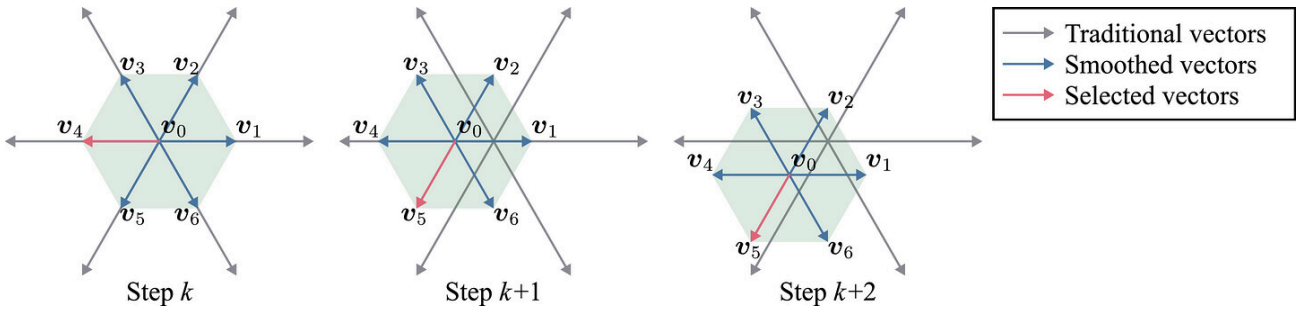


FIGURE 6

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Voltage-smoothing method proposed in [27, 41].

3.2 Long Horizon Prediction and Simplification

the candidate-set size grows exponentially with the increase of prediction horizon. [26] points out this problem. Therefore, most research still employs single-step prediction. However, researchers increasingly recognize that such single-step methods fail to fully exploit the system's performance potential. [57] approximates the PMSM as a double-integrator system and employs dynamic programming to solve the time-optimal control problem. By extracting the optimal trajectory and the associated cost function structure, it successfully overcomes the limitations of single-step prediction when handling large-inertia dynamics. Notably, this exploitation of geometric characteristics originates from optimal control theory and has also been applied to MPTC [58] and cascaded speed control [12].

In [59], an FPGA-based parallel and scalable sphere decoding algorithm (SDA) is developed to solve long-horizon optimization problems efficiently, though with a high hardware cost. [60] proposes an artificial neural network (ANN)-based SDA, achieving a long-horizon prediction on low-cost microprocessors with a lightweight computational burden. Furthermore, [61] comparatively analyses effects between forward time exploration (FTE) and backward time exploration (BTE). Overall, improvement in optimization algorithm is mainly implemented with the aid of SDA [62, 63] and neural network training [64, 65].

However, high-performance optimization algorithms may still require considerable computational resources, thus a number of researchers are focusing on second-step candidate set construction and second-step optimal vector selection. [27] introduces second-step prediction into MPDSC. However, the basic vector for the second step is chosen to be equal to that for the first step, which fundamentally contradicts the core principle of long-horizon prediction strategies. Some researchers construct the second-step candidate set by enumerating all possible switching states from the first step, with at most one switching action [25, 53-56], thereby reducing the number of two-step switching combinations from 64 to 16. [66] introduces a “Hall of Fame” mechanism to store high-quality control actions identified for the second time step, and then promotes them in subsequent control via an incentive term in the cost function. In [67], the control sequence is approximated using Laguerre orthogonal basis functions, significantly reducing the computational burden and achieving long-horizon direct speed control. However, in the above methods, the long-horizon prediction often interferes with the first-step control action. [40] innovatively assigns different weights to the errors at different instants in the cost function to strengthen the influence of the first-step control action, thereby solving this problem.

3.3 Cost Function Form Optimization

Under a basic cost function, the system often exhibits relatively large current harmonics and cannot satisfactorily address the multi-objective optimization of mechanical and electrical performance indices [68]. Related research includes cost term design and cost function structural optimization.

steady-state speed error without significantly altering the cost function form. As stated in [70], interior permanent-magnet synchronous motors (IPMSMs) require special treatment. For IPMSMs, the MTPA strategy is widely used. [56, 71] incorporate the MTPA strategy adding the corresponding term to the cost function, with the aim of maximizing the ratio between the produced torque and the stator current under a given torque demand, so that the required torque can be generated with minimal current.

Although adding cost terms enables more optimization objectives, the main performance index is often weakened, and the tuning of weighting factors becomes very difficult. The cost function designed in [48] includes almost all tuneable performance indices, but selecting five weighting factors is highly challenging. Some methods are adopted in Section 6. [28, 72] point out that the voltage term often inherently contains both speed and current information, and its intrinsic equivalent coefficient can also be interpreted as a form of normalization. In essence, this falls into the scope of DB-MPDSC discussed in the next section.

Cost function structural optimization can alleviate this problem to some extent. In [68, 69], the d- and q-axis current errors are merged into a single term, namely, the flux-error term. [73] develops a cascaded cost function scheme as shown in Figure 7. The first cost function evaluates the torque term and passes three candidate optimal vectors to the second stage, whereas the second cost function includes the speed term, based on which the optimal vector is selected. This approach distinguishes primary and secondary objectives while simplifying factor tuning.

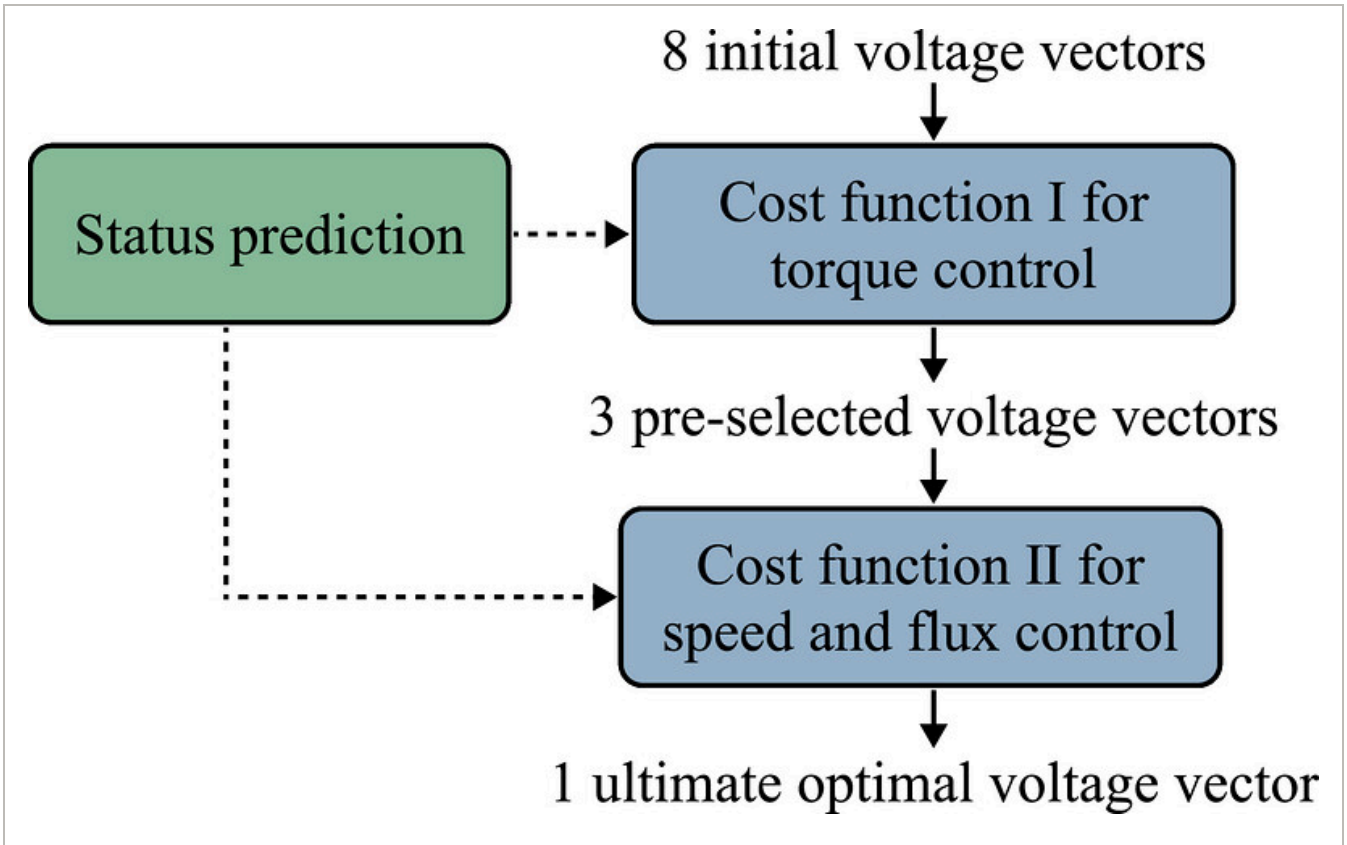


FIGURE 7

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Classical FCS-MPDSC approaches are listed in Table 2, and different terms in the cost function are listed in Table 3.

TABLE 2. Comparative analysis of classical FCS-MPDSC approaches [25-27, 40, 48, 49, 51, 55, 69, 74].

Ref.	Predictive horizon	Cost function	Number of candidate vectors in first step	Number of candidate vectors in second step	Number of switching combinations	Parameter update strategy
[25]	n	$C_\omega + C_{id} + C_{i\text{lim}} + C_{u\text{lim}}$	4	4	4^n	Luenberger observer
[26]	2	$C_\omega + C_{id} + C_{iq} + C_{i\text{lim}} + C_{\text{oth}}$	27	27	729	Luenberger observer
[27]	2	$C_\omega + C_{id} + C_{i\text{lim}}$	8	1	8	Luenberger observer
[40]	3	$C_\omega + C_{id} + C_{iq} + C_{i\text{lim}}$	8	1	8	Sliding-mode observer
[48]	1	$C_\omega + C_{id} + C_{i\text{lim}} + C_{\text{oth}}$	8	—	8	(Not mentioned)
[49]	1	$C_\omega + C_{id} + C_{iq}$	8	—	8	Extended state observer

TABLE 3. Cost function terms used in FCS-MPDSC.

Cost function term	Typical formula	Description
Speed tracking term	$C_\omega = \lambda_\omega(\omega^* - \widehat{\omega})^2$	λ_ω is a weighting factor.ω^* denotes the speed reference.$\widehat{\omega}$ denotes the predicted speed.

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d-axis current tracking term	$C_{id} = \lambda_{id}\left(i_d^* - \hat{i}_d\right)^2$	λ_{id} is a weighting factor.i_d^* denotes the current reference in d-axis.$\hat{i}_d$ denotes the predicted current in d-axis.
q-axis current tracking term	$C_{iq} = \lambda_{iq}\left(i_q^* - \hat{i}_q\right)^2$	λ_{iq} is a weighting factor.i_q^* denotes the current reference in q-axis.$\hat{i}_q$ denotes the predicted current in q-axis.
Current limiting term	$C_{i\text{lim}} = \begin{cases} +\infty, & \text{if } \sqrt{\hat{i}_d^2 + \hat{i}_q^2} > I_{\max} \\ 0, & \text{otherwise} \end{cases}$	$I_{\max}$ denotes the current limitation. **This term prevents current from exceeding the limit**.
Voltage limiting term	$C_{u\text{lim}} = \begin{cases} \xi^2, & \text{if } \xi < 0 \\ 0, & \text{otherwise} \end{cases}$, where $\xi = \sqrt{\left(L_q\hat{i}_q\right)^2 + \left(L_d\hat{i}_d + \psi\right)^2} - \frac{\zeta U_{\text{dc}}}{\sqrt{3}}$	ψ denotes the flux linkage.ζ is a safety factor, with a range of 0 to 1. U_{dc} denotes the DC bus

4 DB-MPDSC

4.1 Reference Voltage Generation

As for DB-MPDSC, generating the reference voltage vector is a key step that bridges the control objectives and the modulation implementation. Accordingly, recent studies have mainly focused on improving discretization accuracy, compensating one-step delay, handling constraints and designing multi-objective optimal reference voltages in modulation.

4.1.1 Discretization Methods and Errors

DB-MPDSC calculates the required voltage vector to achieve precise current tracking within one control cycle by solving the mathematical model of the PMSM. The accuracy of this mathematical model directly impacts the calculation result and, consequently, the overall control performance. In addition to parameter-mismatch effects, which will be further analysed in Section 6, discretization errors arise when applying numerical methods to solve differential equations via difference equations.

Conventionally, the Euler method has been widely adopted due to its computational simplicity, though it introduces a global discretization error of $\mathcal{O}(T_{\text{s}})$, where T_{s} denotes the control period [75]. It is necessary to emphasize that T_{s} should be sufficiently small to guarantee the discretization stability.

To mitigate the accuracy limitations inherent in the Euler method, subsequent studies [28, 48, 76, 77] employ the modified Euler method, also known as Heun's method, further reducing the global error to $\mathcal{O}(T_{\text{s}}^2)$ [75]. This numerical method incorporates a predictor–corrector

more accurate estimation for the next time step.

Alternatively, recent studies [78, 79] adopt a second-order Taylor expansion for discretization, demonstrating enhanced treatment of coupling terms present in the system of differential equations.

From the scope of digital signal processing, several researchers [80] utilize the bilinear transformation method to effectively address the stability problem arisen when the control period T_{s} is not sufficiently small compared with the electrical time constant of PMSM.

4.1.2 One-Step Delay Compensation

During each control cycle, the PMSM control system executes a sequence of operations: sampling mechanical and electrical parameters, computing reference voltages and updating the switching signals for inverter. Within the control loop illustrated in Figure 4, these digital processing stages inherently introduce latency due to their finite execution time.

Since the inverter applies the synthesized switching signals at the beginning of the subsequent control cycle, the processing latency can be quantified as one control period, equivalent to a one-step delay in the discrete-time domain. Alternatively, from the perspective of current tracking, this latency is also termed as “two-step trailing effect” in some studies [81].

Conventionally, this one-step delay is directly compensated by another forward prediction using the discrete-time mathematical model of the PMSM. Specifically, the reference voltage computed during the $k\mathrm{th}$ cycle is applied during the $(k+1)\mathrm{th}$ cycle, and influences the electrical performance sampled during the $(k+2)\mathrm{th}$ cycle. To account for this timing sequence, the current for the $(k+2)\mathrm{th}$ cycle must be predicted during the $k\mathrm{th}$ cycle, and become the controlling object. Thus, in addition to the initial prediction from $k\mathrm{th}$ cycle to $(k+1)\mathrm{th}$ cycle, a second prediction, from $(k+1)\mathrm{th}$ cycle to $(k+2)\mathrm{th}$ cycle, is performed to compensate the one-step delay [82, 83].

However, compared with one-step prediction, two-step prediction inherently accumulates errors due to lumped disturbances. To suppress these disturbances and enhance accuracy, recent studies [84, 85] have replaced conventional model-based prediction methods with sliding-mode disturbance observers.

Alternatively, other research [81] addresses the system latency by optimizing the time schedule of sampling and updating, thereby improving tracking performance without relying on extensive hardware requirements.

4.1.3 Constraint Problem

Since the cost function of DB-MPDSC does not include constraints, the handling of constraint issues requires additional processing. [28] proposes a unified limiting circle, in which the current limit is converted into a voltage limiting circle on the α–β plane. The synthesized vector of the

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computational burden but insufficient accuracy. [30] introduces a reference-current design targeting the “minimum-current operating point”. More commonly, a limiting-circle method combined with a magnitude-scaling strategy is adopted [86], that is, the direction of the voltage vector is kept unchanged while its magnitude is adjusted so that it falls within the limiting circle.

However, if the speed cannot perfectly track the reference at instant $k+2$ due to various limitations, the actual control performance obtained from the above objective will deteriorate. Shown in Figure 8, [87] points out that there exists a maximum speed increment within one time-step due to the limitation of maximum torque. The speed error segment beyond this maximum speed, referred to as invalid error in Figure 8, does not contribute to voltage-vector selection, thus should not be assigned any weight in the cost function. This study further indicates that this is the fundamental reason for problems associated with conventional short-horizon prediction.

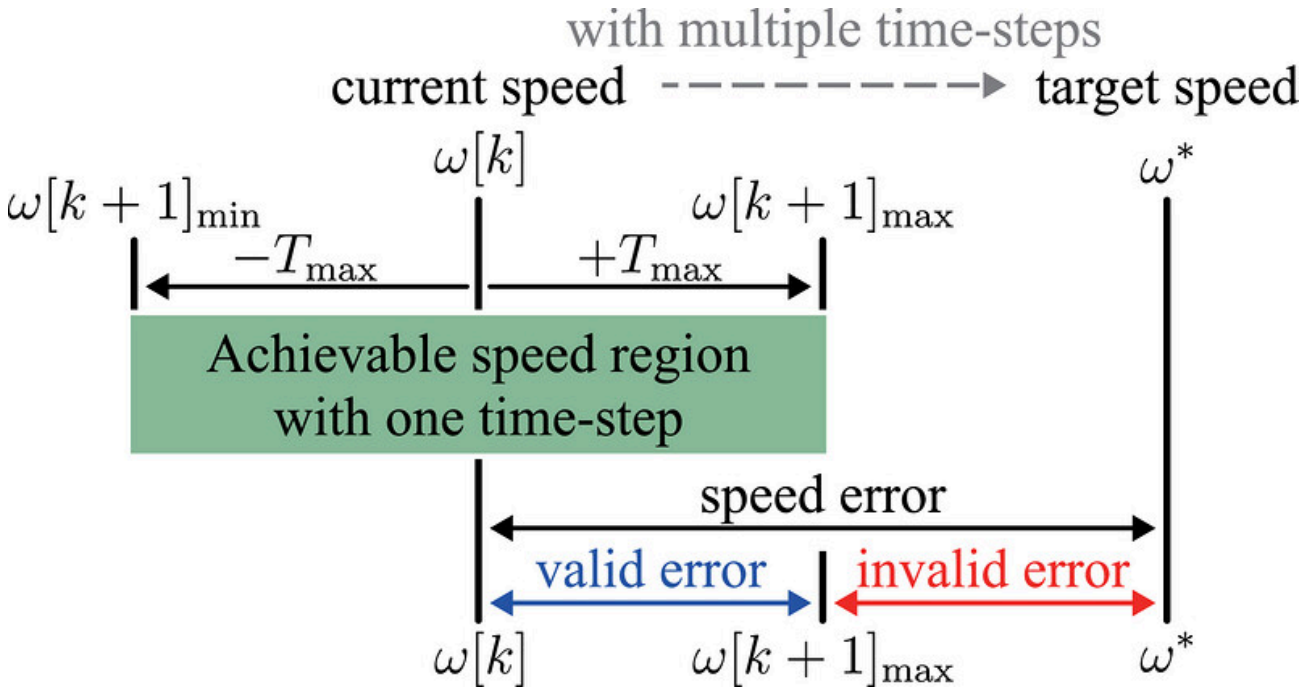


FIGURE 8

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Reconstruction of the speed error [87].

4.2 Switching Signal Modulation

Compared with conventional methods, DB-MPDSC imposes more stringent requirements on the modulation scheme. The reference voltage-vector trajectory must be reproduced as accurately as possible within a finite carrier period to maintain the fast dynamic performance of speed and

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4.2.1 Single-Vector Methods

Single-vector methods adopt the finite-control-set concept by selecting the optimal basic vector from the set of basic vectors and applying it over the entire period. [89] analytically derives the unconstrained optimal voltage vector by calculating the partial derivative of a dual-objective cost function. Subsequently, the optimal basic vector is selected following constraint processing. [90] uses the Manhattan distance between the candidate vector and the optimal vector as the cost function to obtain the optimal basic vector. However, there are many problems with the control performance under single-vector operation, whereas the integrated speed-current architecture leaves sufficient room for some complex algorithms. On this basis, many studies draw on current loop control in a cascaded structure and use SPWM [27, 41] or SVPWM [86, 91] to generate the switching signals.

4.2.2 Multi-Vector Methods

Multi-vector methods apply multiple vectors within one control period. [73] and [92] suggest that ripples in torque, speed or flux are reduced by inserting a zero-voltage vector. [93] explores the optimal combination of any two vectors and the duty-ratio optimization strategy. [76] proposes calculating the duties of selected vectors according to the deadbeat principle based on the d–q axis reference voltages. In [94], the duty ratios of two vectors are computed by solving an online constrained quadratic optimization problem, so as to minimize the weighted squared error. In addition to two-vector methods, three-vector methods have also been extensively studied [95, 96]. However, any three nonzero vectors in a plane are necessarily linearly dependent. Compared with two-vector methods, this incurs higher switching losses and also makes duty-ratio computation more difficult. Therefore, so-called three-vector methods usually consist of two nonzero vectors and one zero vector. [97] provides a basic solution procedure: first determine the sector to which the target vector belongs, select the voltage vector at the centre of that sector as the first voltage vector, choose an adjacent voltage vector to the first one that can further reduce the cost function as the second voltage vector, and finally select the zero vector as the third voltage vector. Virtual vectors are essentially also a multi-vector method. They avoid the online duty-ratio computation step, because the corresponding basic vectors and duty-ratio information are already obtained during virtual-vector generation. It is advantageous in terms of online computational burden [44]. [45] constructs the virtual vector set using two active vectors and one zero vector. [38] proposes a virtual-vector-based MPDSC control paradigm, enabling efficient vector synthesis through virtual-vector construction and voltage-vector pre-selection.

4.2.3 Modulation Methods

Introducing switching-signal modulation is beneficial for suppressing current harmonics. In addition to SVPWM/SPWM methods, a commonly used modulation scheme employs a symmetric seven-segment PWM sequence [44, 94] as shown in Figure 9. In [98], a tracking term

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For low-inductance machines, [99] adds a bus circuit composed of high-frequency switching devices at the front end of a conventional inverter, significantly suppressing current ripple. In addition, some intelligent modulation methods have been reported [100], but lacking practical value in industrial real-time scenarios.

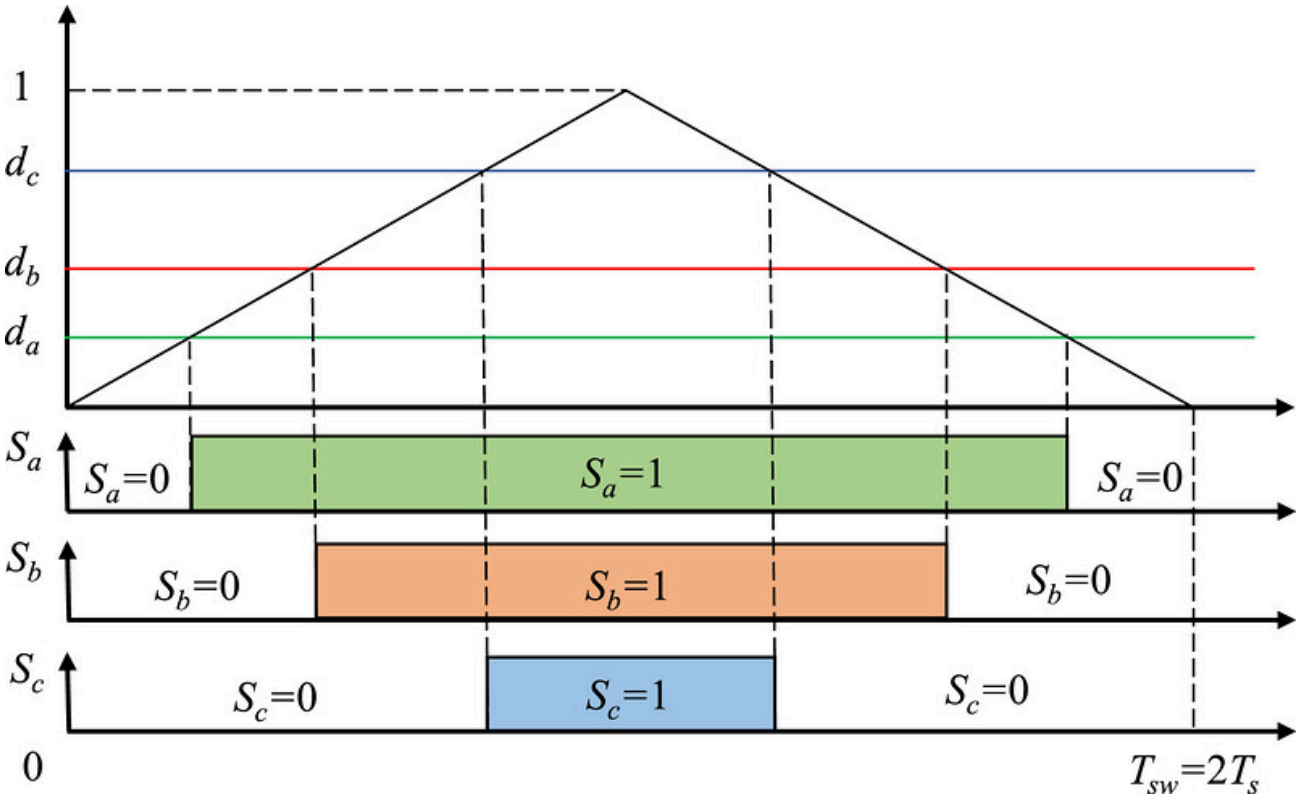


FIGURE 9

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A commonly used modulation scheme applied in [94].

Classical DB-MPDSC approaches are listed in Table 4. u^* denotes the reference voltage, and u_i denotes a candidate voltage vector. $\|u^* - u_i\|_2$ represents the L2 norm for $u^* - u_i$, namely the Euclidean distance, whereas $\|u^* - u_i\|_1$ represents the L1 norm for $u^* - u_i$, namely the Manhattan distance.

TABLE 4. Comparative analysis of classical DB-MPDSC approaches [28-30, 38, 76, 77, 90, 91, 101, 102].

Ref.	Discretization	Cost function	Voltage vector and modulation method	Disturbance compensation metho
[28]	Second-order	$\|u^* - u_i\|_2$	Two active vectors with variable duty cycle	Sliding-mode observer

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[29, 30]	First-order	(Not used)	SVPWM	Extended state observer
[38]	First-order	$\|u^* - u_i\|_2$	One virtual vector, generated by fundamental vectors with specific duty cycles	Kalman observer
[76]	Second-order	$\|u^* - u_i\|_2$	One active vector and one zero vector, with variable duty cycle	(Not mentioned)
[77]	Second-order	(Not used)	SVPWM	Adaptive parameter identification
[90]	Second-order	$\|u^* - u_i\|_1$	(Not mentioned)	Super-twisting sliding-mode observer
[91]	(Not mentioned)	(Not used)	SVPWM	Extended state observer
[101]	First-order	$\|u^*$	Two active vectors with variable duty	Sliding mode observer

5 CCS-MPDSC and EMP-DSC

5.1 Modelling

5.1.1 Model Simplification for CCS-MPDSC

Directly extending CCS-MPCC to CCS-MPDSC presents severe challenges regarding the solution of the online optimization problem. The system is characterized by strong multi-variable coupling and the coexistence of multiple constraints. Consequently, complex solution processes can degrade the system's real-time performance.

A significant pathway involves compressing the prediction horizon through model order reduction. [33] introduces an equivalent speed tracking error to reduce the original second-order system to a first-order one, enabling effective prediction of speed dynamics even within a short prediction horizon. By adopting an incremental model, the nonlinear amplitude constraints on current and voltage are converted into linear constraints on incremental voltage. Additionally, the Hildreth algorithm is utilized to achieve sub-microsecond solution times.

In recent years, the introduction of Laguerre functions has provided a breakthrough solution for implementing long prediction horizons. [67, 103, 104] utilize Laguerre orthogonal basis functions to parameterize future control trajectories. This technique reduces the dimensionality of decision variables from the length of the control horizon to the number of Laguerre coefficients, thus rendering online optimization with long prediction horizons feasible. [67] further combines Laguerre functions with the aforementioned reduced-order model.

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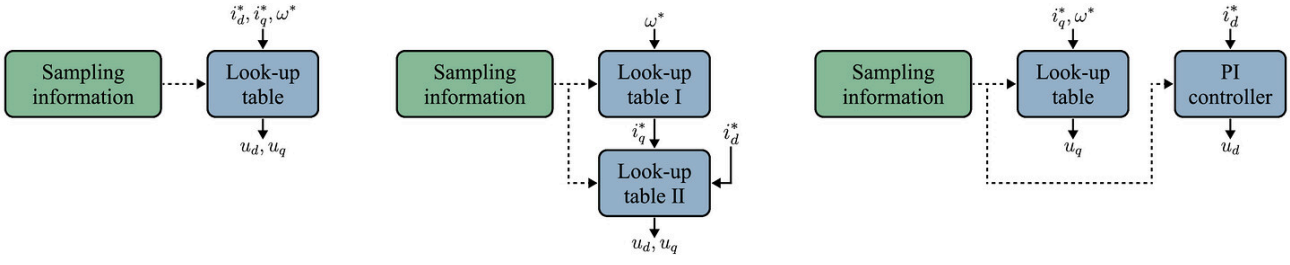
Other methods have also been proposed. [105] introduces particle swarm optimization (PSO) into the CCS-MPDSC framework. By utilizing the cost function as the fitness function for PSO, this approach circumvents the complex computations associated with traditional QP solvers, approximating the optimal solution through an iterative particle swarm search within the continuous voltage space. Moreover, [32] proposes a modulated MPDSC based on a voltage-type cost function, improving current quality while maintaining a rapid dynamic response.

5.1.2 State Vector Design

Within the basic framework, the construction of the state vector and the selection of cost terms are investigated. For MPCC, the state-vector dimension is relatively low, which is beneficial for multi-objective optimization [106]. However, with more objectives involved, the state-vector design must balance control performance and dimensionality simultaneously [107]. The cost function depends on the state variables and the input sequence. Therefore, when reformulating the problem into a state-space form, the state vector should be constructed according to the cost terms, whereas the control inputs are chosen as the d- and q-axis voltages. The state vector defined in [108, 109] exploits a property of SPMSMs that $i_d^* = 0$, and proposes neglecting the coupling term $\omega_m i_d$ to reduce the state-space dimension. [110] follows this idea and further incorporates the torque term into the state vector. Combined with a Kalman observer, torque ripple is reduced. Building on this, [111] replaces the forward Euler discretization with a second-order Taylor expansion to improve discretization accuracy. In contrast, [112] retains the coupling term $\omega_m i_d$ to improve accuracy, at the cost of increasing the state-vector dimension to nine. [113] also retains $\omega_m i_d$, while treating T_L as a measured constant disturbance and adding a constant term to the state vector.

5.1.3 Structures for EMP-DSC

As discussed above, the dimensionality required by the basic control objectives is high, leaving limited freedom for the state vector design. Therefore, further improvements should be pursued at the architectural level, partly shown as Figure 10.



(a) Conventional structure [108]. (b) Cascaded structure [114]. (c) Dual structure [38].

FIGURE 10

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To address the coupling and excessively high dimensionality in conventional EMP-DSC, [114] proposes a cascaded architecture comprising a speed loop EMP controller and a current loop EMP controller. By contrast, [115, 116] adopt a dual closed-loop strategy, where the u_d is generated by a PI controller and a torque term is incorporated, thereby reducing the state vector to five dimensions. Compared with [114], this dual loop strategy requires less computational resources and can achieve a larger bandwidth. However, it does not account for the coupling terms.

5.2 Constraint Optimization

5.2.1 CCS-MPDSC

Constraint optimization strategies combine robustness with safety, focusing on obtaining secure and feasible solutions. [117] addresses optimization infeasibility and sensitivity to disturbances encountered by traditional MPC. By embedding recursive feasibility and L2 gain as hard constraints within the optimization problem, this study utilizes linear matrix inequalities (LMIs) to transform the problem into a convex optimization formulation. This method theoretically guarantees that the controller possesses a solution under all operating conditions and exhibits a specified capability for suppressing external disturbances. [118] introduces the control barrier function (CBF) into MPDSC to resolve the fundamental trade-off between current safety hard constraints and dynamic performance, with innovation lying in the construction of a robust adaptive CBF (RACBF). By employing a preset performance function to dynamically compensate for estimation errors, conservative design is avoided. Simultaneously, an adaptive mechanism dynamically adjusts the barrier strength based on the proximity of the current to its limits. The RACBF constraint is embedded within a minimum-intervention QP problem that possesses an explicit solution, resulting in an extremely low computational burden. This provides a novel paradigm for handling system hard constraints that balances strictness with flexibility.

5.2.2 EMP-DSC

Conventional EMP-DSC methods exhibit limitations in constraint handling. Early research attempts to design the optimization problem with the aid of geometry. To address current ripples and flux-weakening constraints, [119] maps the optimal control problem onto the current plane, converting the control problem into a geometric projection problem that seeks the point within an elliptical feasible region closest to the unconstrained solution. By designing a fast projection algorithm based on Newton's iteration, the complex nonlinear programming problem is simplified into intuitive geometric calculations, significantly enhancing solution efficiency.

Region constraints typically lead to a large number of PWA regions. To this end, [120] proposes a line-constraint method that converts the conventional two-dimensional current and voltage region constraints into one-dimensional line constraints, significantly reducing the number of piecewise-affine regions. In addition, traditional EMP-DSC commonly uses a regular hexagon to

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constructs an irregular hexagonal constraint that employs more line segments to approximate the circular boundary in critical operating quadrants as shown in Figure 11, thereby enlarging the reachable set of voltage and current vectors. Furthermore, an optimal-trajectory flux-weakening strategy is designed such that the d–q axis currents can be adaptively coordinated along the linear constraints, satisfying both constraints while balancing speed tracking and load sharing, thus improving full-speed-range performance.

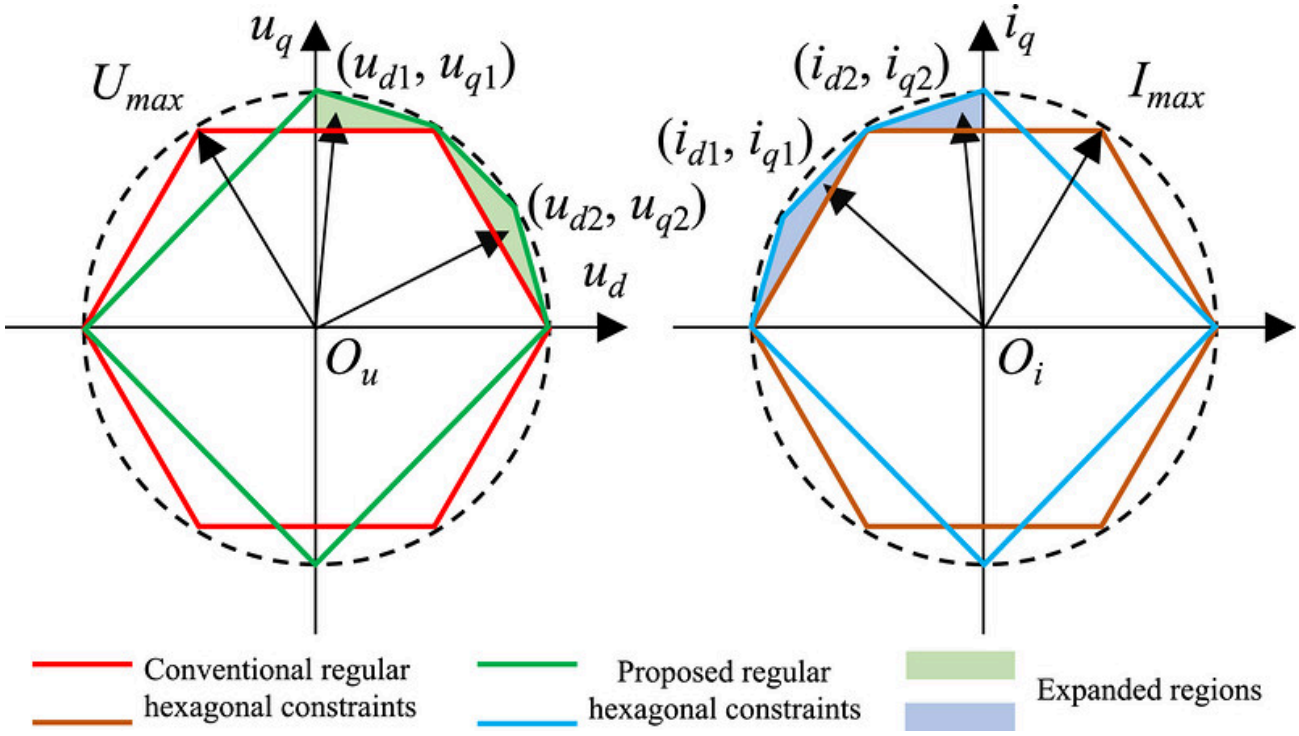


FIGURE 11

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The irregular hexagonal constraints proposed in [34].

5.3 Compensation Enhancement

5.3.1 CCS-MPDSC

Strategies for predictive model enhancement based on compensation do not directly alter the solution algorithm. Instead, they employ observers to estimate and compensate for internal and external system disturbances in real time. It is noteworthy that the compensation enhancement methods discussed herein are typically integrated within the optimization problem, whereas the approaches introduced in Section 6 operate in parallel with the optimization process. [31] incorporates a disturbance observer (DOB) into the single-loop MPC framework, embedding the estimated lumped disturbances of both current and speed loops into the discrete prediction model. [121] defines “lumped exogenous signals” from the perspective of output regulation. By

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the optimization solution shifts from statically tracking a reference point to dynamically tracking an optimal trajectory. Based on the reduced-order model of PMSMs, [122] adopts an ultra-local model (ULM) to replace the physical model and estimates lumped disturbances online. Combined with the proposed online adaptive mechanism (OAM), dynamic adjustment of closed-loop poles is achieved, maintaining stability even under significant parameter perturbations. To address the periodic disturbances common in servo systems, [123] proposes a dual-layer MPC framework based on an iterative learning observer (ILO). The upper-layer MPC precisely calculates the reference trajectories for states and controls based on this information, whereas the lower-layer MPC tracks these trajectories, thereby realizing the estimation and preview of disturbance sequences across the entire prediction horizon.

5.3.2 EMP-DSC

Since the EMP-DSC optimization problem is solved offline based on a nominal model, dynamic disturbances must be compensated online through real-time adjustment of the control action. Such disturbances include parameter mismatches and multi-source external disturbances. The extended state observer (ESO) treats internal and external disturbances as a lumped disturbance for estimation and compensation [124]. The associated ultra-local model (ULM) has also attracted increasing attention in motor control in recent years [125, 126], resulting in an active research topic on ULM-EMP-DSC [35, 36]. ULM-EMP-DSC emphasizes data-driven design of dynamic control inputs and shares certain similarities with model-free control. Problems related to lumped disturbance estimation will be further discussed in Section 6.

Classical CCS-MPDSC and EMP-DSC approaches are listed in Tables 5 and 6.

TABLE 5. Comparative analysis of classical CCS-MPDSC approaches [31, 33, 67, 103-105, 117, 118, 121, 122].

Ref.	Core methodology	Prediction model	Disturbance handling	Constraint handling	Computational burden optimization	Stability analysis
[31]	Composite single-loop MPC + DOB	Simplified model+ DOB estimation	Second-order linear DOB	Current and voltage constraints via Lagrange duality + Hildreth QP	Standard QP	Observer error convergence analysis, no closed-loop proof
[33]	Short-horizon CCS-MPDSC	Reduced-order model	Incremental model	Current and voltage	Short horizon + Hildreth's QP	Not explicitly proved

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					optimization	
[67]	Laguerre-based long-horizon MPC	Augmented incremental model	Incremental model	constraints Hard via QP constraints via Hildreth's QP	Laguerre functions	Closed-loop poles analysis

TABLE 6. Comparative analysis of classical EMP-DSC approaches [34-36, 110, 111, 115].

Ref	Control structure	State vector dimension	Parameter update strategy	Innovation
[34]	Conventional form	10	Extended state observer	Expanding limit boundary
[35]	Conventional form	9	Extended state observer	Applying active disturbance rejection control
[36]	Conventional form	9	Data-driven observer	Data-driven approach
[110]	Conventional form	8	Kalman observer	Adding load torque to model
[111]	Conventional form	8	Kalman observer	Discretization by second-order Taylor expansion
[115]	Dual form	3	(Not mentioned)	Dimensionality reduction of states

6 Weighting Factors

Compared with MPCC, MPDSC involves a greater number of optimization objectives, making the design of weighting factors more challenging. The selection of weighting factors directly influences the trade-off among various control objectives, thereby determining the system's dynamic and steady-state performance. Existing research generally falls into two categories: weighting factor tuning and weighting factor elimination.

6.1 Weighting Factor Tuning

6.1.1 Algebraic Methods

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current term and the speed term. By fixing the weighting factor of one term, only the factor of the other requires adjustment [27]. Notably, the weighting factor for the speed term must be significantly larger than that of the current term, with the priority to speed tracking. For multi-objective optimization, algebraic methods primarily encompass adaptive weight design [32, 48, 49, 55, 127, 128], quadratic stability design [79, 129] and enhanced weighting stability [33, 117, 122].

Regarding adaptive weight design, researchers have proposed dynamic adjustment strategies tailored to different control objectives. [55] demonstrates that increasing the torque term weight during steady-state operation effectively reduces torque ripple and improves current quality. To target the speed error term and the high-frequency component in torque, [48] designs an exponential dynamic weighting factor varying with the speed error. [49] utilizes the speed error as a dynamic weighting factor and introduces distinct q-axis current references for transient and steady states. These methods have collectively enhanced both dynamic and steady-state control performance. Additionally, [32] incorporates an integrator loop outside the MPC framework to adjust the speed term weight based on speed error in real-time, thereby eliminating steady-state errors. Offline design methods have also been applied: [127] performs offline tuning of weights using PSO, whereas [128] employs fuzzy inference rules to construct fuzzy sets for the dynamic update of weighting coefficients.

Regarding quadratic stability design, certain research utilizes the quadratic cost function as a Lyapunov function. By employing LMIs to solve offline for a weighting factor matrix that satisfies stability and rapidity constraints, dynamic response speed is improved while guaranteeing system stability [129]. Similarly, replacing the traditional weighted sum cost function with an offline-solved quadratic cost function allows for the optimized design of weighting coefficients [79].

Regarding enhanced weighting stability, Laguerre functions are introduced to parameterize the control trajectory, transforming decision variables from high-dimensional control sequences into low-dimensional coefficient vectors. This transformation significantly reduces the dimension of the Hessian matrix in the optimization problem, indirectly improving the numerical stability of the weighting matrix and rendering the system less sensitive to weight variations [33, 117]. Building on this, [122] further introduces exponential decay weighting to improve numerical stability over long prediction horizons.

6.1.2 Network-Based Methods

Network-based methods derive optimal weighting coefficients through complex procedures such as offline network training and optimization. Unlike traditional methods relying on precise mathematical models, network-based methods employ a data-driven paradigm to uncover the complex mapping relationships among system performance indices, weighting factors and operating conditions, thereby facilitating the objective design and dynamic adjustment of

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Early research primarily focuses on utilizing AI to establish offline surrogate models of the system. The non-dominated sorting genetic algorithm II (NSGA-II) has been widely applied to offline optimization. By generating Pareto-optimal front solutions, it systematically balances conflicting objectives such as torque ripple and switching frequency, providing an objective basis for selection across different application scenarios [130, 131].

To achieve real-time adjustment, ANNs have been introduced for online inference. Researchers have deployed ANNs on FPGA platforms, utilizing integer quantization techniques to achieve low-latency inference and output optimal weighting factors in real-time according to operating conditions [132]. To further enhance accuracy, genetic algorithms (GA) have been employed as a back-propagation tool. By combining the fitting capability of ANNs with the global search advantage of GA, optimal weights under various operating conditions can be accurately predicted [130]. Addressing the issue of adaptability to multiple operating conditions, a cascaded network structure has been proposed. The lower-layer network establishes a high-precision surrogate model offline, whereas the upper-layer network performs online inference, achieving dynamic weight adjustment with an extremely low computational burden [133].

Deep learning technologies have further enhanced network representation capabilities. Residual networks (ResNet) utilize residual connections to resolve the vanishing gradient problem in deep networks, thereby improving the accuracy of weight prediction [134]. Aggregated ResNet introduces the concept of “cardinality”, simplifying parameter tuning and enhancing network robustness to motor parameter variations [135]. Simultaneously, to address the problem of high online computational complexity, [135] implements ResNet as a real-time executable look-up table (LUT). The simulated annealing algorithm offers a training-free online search scheme, rapidly converging to optimal weights using the error area as a criterion [136]. Regarding the decision-making problem following multi-objective optimization, ranking-based decision methods have been proven to outperform methods such as Euclidean distance in terms of comprehensive performance, providing theoretical support for selecting the final weights from the Pareto solution set [137].

6.2 Weighting Factor Elimination

The elimination of weighting factors is primarily categorized into complete elimination [40, 105], weighting factor simplification [51, 73] and optimization problem simplification [103, 118-120].

In terms of complete elimination, certain research attempts to reconstruct the cost function to entirely discard weighting factors. [40] formulates the cost function by directly summing the speed error and flux error, effectively eliminating the weighting factor while assigning higher coefficients to the near-term horizon. Another approach involves directly seeking the optimal voltage vector via heuristic search. For instance, by introducing the PSO algorithm and utilizing the cost function as the fitness function, the weighting matrix inherent to traditional QP solutions is completely bypassed, thereby essentially achieving weighting factor elimination

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Weighting factor simplification refers to reducing the number of tuneable weighting factors or the dimensionality of the weighting matrix for multi-objective optimization problems. [40] merges the d- and q-axis current errors into flux error, which inherently assigns natural weights to the d- and q-axis current terms. [108, 109] propose neglecting coupling terms for SPMSMs to reduce the state-space dimensionality, thereby subsequently reducing the dimensionality of the weighting matrix. Additionally, simplification can be achieved through structural design of the cost function. [73] designs a cascaded cost function, decomposing the multi-objective optimization into two separate problems, with only one weighting factor in the second optimization step to be tuned. [51] introduces a sliding mode surface term into the cost function, transforming the coordinated control of speed and torque into a convergence problem of a single sliding mode surface, thereby avoiding the complex weighting factor design.

Regarding optimization problem simplification, researchers employ constraints or transform the optimization problem formulation, such that the role of weighting factors is superseded by other mechanisms. [118] transforms current safety constraints into dynamic RACBF constraints. Although a QP problem still needs to be solved, the dominant factor shifts from performance weighting to the self-tuning mechanism of safety parameters. [119] simplifies the multi-objective optimization problem into a geometric projection problem, eliminating the need for weighting trade-offs between different control objectives. Furthermore, [103] integrates the L2 gain criterion into an infinite horizon cost function. By utilizing LMIs to transform the optimization problem into a convex optimization form, the selection of weighting factors is converted into an LMI feasibility problem. [120] transforms the traditional circular current amplitude constraint into a linear constraint, with weighting factor not affecting the location for closed-loop poles, thereby simplifying the weighting factor design.

7 Robust problem

7.1 Analysis of Parameter-Mismatch Effects

The electrical and mechanical parameters of PMSM drive systems often exhibit time-varying drifts, resulting in parameter mismatches between the actual plant and its ideal model. Common mismatch phenomena are listed in Figure 12. Parameter mismatches not only degrade the capability to predict future behaviour, but also increase steady-state errors and deteriorate transient performance, and may even challenge the closed-loop stability in extreme cases [138].

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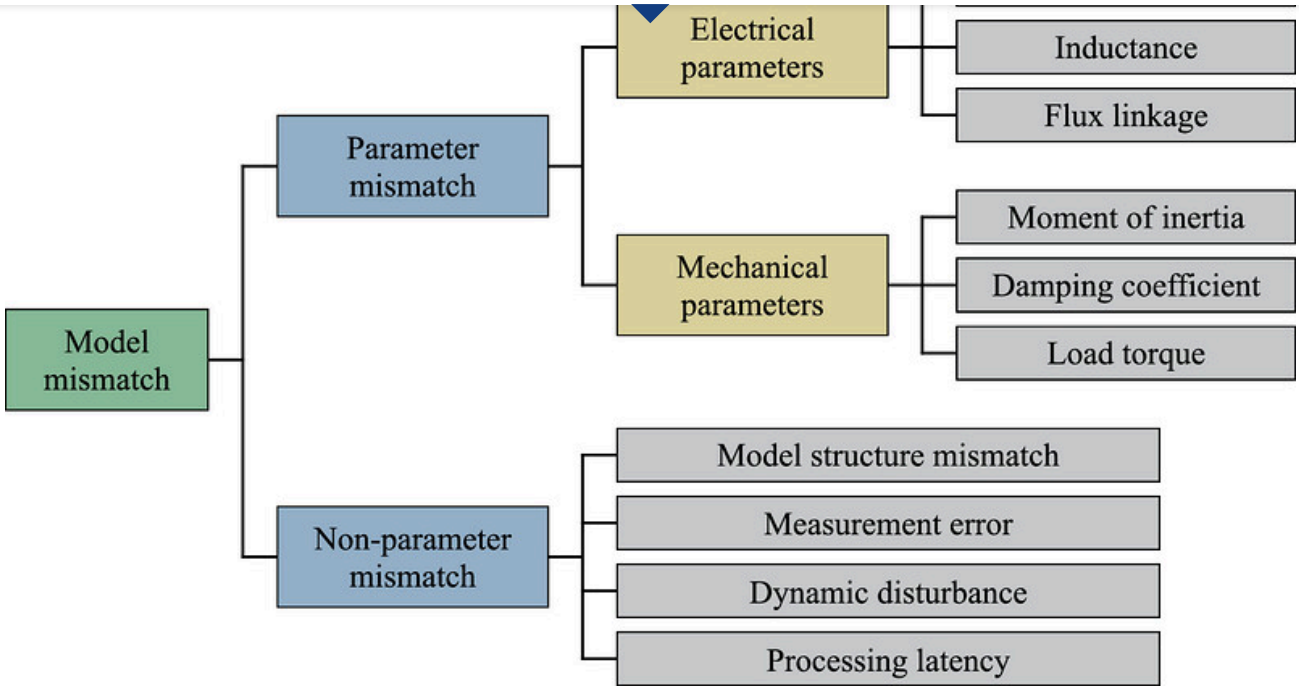


FIGURE 12

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Robustness problems of the predictive model.

Although the parameter mismatch problem has been widely acknowledged, few studies provide a rigorous theoretical analysis. Focusing on electrical parameters, [139] reports that the impact of inductance errors is asymmetric: the degradation becomes more pronounced when the actual inductance is smaller than the modelled one. In addition, resistance errors mainly affect steady-state accuracy, whereas inductance errors affect both steady-state and dynamic performance. [140] proposes a visual analysis method using a Voronoi diagram, showing that flux-linkage mismatch can cause complex shifts of the reference voltage vector in the α–β plane.

In [101], parameter errors are directly mapped to voltage errors and sensitivity curves are derived. In contrast, [141] finds that inertia mismatch has the largest impact in dynamic scenarios. For a sequence MPDSC scheme, [142] shows that it is robust to moderate variations in resistance and inertia, but is sensitive to underestimated inductance and flux-linkage mismatch, with performance degradation becoming particularly evident at high speeds.

7.2 Update Strategies for Electrical Parameters

Electrical parameters include the stator resistance, inductance and flux linkage, and these parameters typically vary slowly. With the continued development of MPCC, observer-based approaches have been extensively investigated [143-145]. However, for MPDSC, the presence of mechanical parameters implies that employing an observer solely for electrical parameters would further increase the computational burden. Therefore, existing solutions are more often

methods [151, 152]. In [149], an online parameter identification method based on MRAS is developed, where the adaptation law is designed using Popov hyperstability theory. [152] employed a variable-step Adaline neural network to build an identification model for the electrical parameters. In [153], a data-driven RLS estimation method with real-time capability is proposed.

In [154] uses high-frequency current injection to enable simultaneous online identification of parameters. [155] proposes a PMSM multi-parameter identification approach based on a high-frequency equivalent-impedance model, applicable to both online and offline scenarios. On this basis, RLS or the high-frequency impedance model is then used to identify the stator resistance and permanent-magnet flux linkage. In [156], an offline identification "networking" procedure together with fast online computation is proposed to reduce the online computational burden.

Online identification of the flux-linkage typically relies on the resistance and inductance estimates, leading to coupling and accumulation of estimation errors. It has been shown that an incremental model can be constructed from the discrete-time PMSM equations at two adjacent sampling instants to eliminate the flux-linkage term [157, 158]. However, incremental-model-based formulations have not yet been sufficiently explored in MPDSC, and are currently more commonly applied in MPCC and MPTC [159].

7.3 Update Strategies for Mechanical Parameters

Mechanical parameters include inertia, damping and load torque. These parameters may experience abrupt changes in practice and therefore must be tracked rapidly. Accordingly, online updating of mechanical parameters has become a major research focus in MPDSC. The damping coefficient is often neglected due to its little influence on the system response.

Mechanical-parameter estimation strategies include RLS [160, 161], model reference adaptive schemes [69, 162], gradient-descent methods [163], Kalman filtering [164], observer-based approaches [28, 79, 165] and intelligent methods [166]. In [167], an MRAS-based approach combined with the Landau iterative algorithm is developed to update the inertia parameter online with fast convergence. Based on an analysis of how variations in different parameters affect the operational stability of the motor, [161] proposes a least-squares method with a forgetting factor to update the rotational inertia online. In [168], a load disturbance observer based on a two-stage extended Kalman filter (EKF) is proposed. A two-stage transformation matrix is employed to decouple the high-dimensional covariance computations inherent in the original EKF, thereby significantly reducing the computational burden. It reflects a widely adopted cascaded structure for mechanical-parameter updating. Common load torque observers include the Luenberger observer [69], sliding-mode observer [28], and Kalman observer [79]. The comparison of different kinds of observers is provided in Table 7.

TABLE 7. Comparison of different kinds of observers.

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Model dependence	High	Low to medium	Medium	Medium to high
Disturbance rejection	Medium	Strong	Very strong	Strong
Noise sensitivity	Medium	Medium to high	High	Low to medium
Computational burden	Low	Medium	Medium to high	Medium to high

Although such decoupled update strategies reduce the computational burden, they may introduce coupling-induced estimation errors. To address this problem, [169] proposes an extended sliding-mode mechanical-parameter observer (ESMMPO) to simultaneously estimate the rotational inertia, viscous friction coefficient and load torque. Furthermore, [170] proposes a dual extended sliding-mode observer (DESMO) for synchronous estimation of the total system inertia and the load torque. By introducing a robust activator, it suppresses the coupling effects between the two estimation errors, achieving strong robustness and low sensitivity.

7.4 Compensation Strategies for Total Disturbance

For the prediction model used in MPDSC, numerous parameters vary dynamically and are coupled with each other. Consequently, the system exhibits compound uncertainties that are difficult to model accurately or identify completely. Therefore, developing compensation strategies oriented toward the “total disturbance” has become an effective approach to improving the robustness of MPDSC. In [91], an active-disturbance-rejection direct speed control method is proposed by incorporating robust ADRC principles. A simplified motor model with time-varying coefficients is developed in [171], employing a gradient-based method with adaptive gain for online parameter identification. In [101, 102], a full-parameter disturbance and load observer (FPLO) is designed, and a sliding-mode-control-based structure is adopted to simultaneously estimate multiple classes of disturbances. Different disturbance observers are used in other research [35, 74, 172].

7.5 Model-Free Strategies

The aforementioned parameter update and disturbance compensation strategies rely on model-based frameworks. In recent years, model-free predictive control (MFPC) has offered new perspectives for the robust design of MPDSC, primarily encompassing ULM-based methods and network-based methods.

7.5.1 ULM-Based Methods

dynamics and external disturbances uniformly, which is subsequently estimated and compensated for by a designed observer. This methodology was initially applied to model-free predictive current control (MFPCC) [120, 173] and has subsequently been investigated in model-free predictive direct speed control (MFPDSC) [34, 36].

At the observer design level, research efforts are dedicated to enhancing the accuracy and dynamic response speed of disturbance estimation. Addressing the limited observation bandwidth of ESOs, [174] proposes second-order and first-order data-driven observers. These designs simplify the structure while maintaining high observation bandwidth. [175] focuses on the performance degradation of nonlinear ESOs under large disturbances. By designing an improved nonlinear function, it achieves rapid and accurate estimation of large-scale disturbances. [176] further equates speed disturbances to current disturbances and introduces a third-order ESO. Through root locus analysis, the study reveals the influence mechanism of input gain on dynamic performance. [177] adopts algebraic parameter identification for disturbance observation and compensation. It achieves cascade-free control by designing an adaptive gain feedback loop, which combines a sliding mode surface with the gradient descent method.

At the control objective level, researchers have expanded model-free methods from pure disturbance compensation to multi-objective synergistic optimization. Building upon disturbance estimation, [175] proposes an algebraic method to directly calculate the duty cycle, thereby ingeniously eliminating weighting factors. [174] addresses the challenge of current constraints in cascade-free structures by designing a q-axis current constraint compensation module. This approach fully utilizes current dynamics while ensuring system safety, further leveraging the advantages of cascade-free control.

Regarding the expansion of application scenarios, model-free methods are beginning to merge with other advanced control technologies. [178] combines the ULM with position sensorless control. It utilizes the $x-y$ subspace for rotor position estimation and employed an ESO to suppress harmonic current interference. This achieves high-precision sensorless operation at low speeds, extending the application of model-free methods to multiphase motor systems.

7.5.2 Network-Based Methods

In recent years, network-based model-free methods have offered new avenues for MPDSC to eliminate parameter dependence and enhance adaptive capabilities. These methods utilize neural networks to learn dynamic characteristics directly from system input-output data.

In [54] is among the early works to adopt a back propagation (BP) neural network to replace the mathematical model of the PMSM. By employing online sliding window learning to track local system dynamics, it achieves speed prediction and FCS optimization, preliminarily validating the adaptive capability of neural networks under parameter perturbations. [179] further proposes replacing the predictive model with an offline-trained feedforward neural network (FNN). The

maintains superior control performance even under parameter mismatch conditions. [180] introduces the echo state network (ESN), whose dynamic reservoir structure is well-suited for time-series prediction. Furthermore, by utilizing the Woodbury matrix identity to enable recursive online updates of the output layer weights, this method significantly reduces the computational burden while ensuring prediction accuracy. [181] integrates a neural network predictor into a data-driven model-free adaptive control (MFAC) framework. It utilizes the NN to estimate unknown nonlinear dynamics, combining the MFAC control law with FCS optimization. By employing offline imitation learning to shift the majority of the online computational burden to the training phase, the study verifies strong robustness against parameter mismatches and measurement noise within a modular multilevel converter (MMC) system.

8 Comparative Analysis

8.1 Simulation and HIL Experiment

Four classical schemes of algorithms from [25, 28, 31, 34] are selected for simulation comparison. Parameters for the motor used in simulation are listed in Table 8. The target speed is set to 1000 rpm. The load torque is initially 6 N m and then steps up to 8 N m at 0.1 s. Three-phase currents and rotational speeds for all four schemes are measured and recorded, shown in Figure 13. To verify robustness, the speed response near load torque stepping are enlarged for better visualization. The total harmonic distortion (THD) of three-phase currents is calculated, shown in Table 9.

TABLE 8. Parameters for the motor used in simulation.

Symbol	Quantity	Value
R	Stator resistance	0.9585 Ω
L_d	d-axis inductance	5.25 mH
L_q	q-axis inductance	5.25 mH
ψ	Flux linkage	0.1827 $\mathrm{V \cdot s}$
p	Pole pairs	4 pairs
J	Rotor inertia	0.0006329 $\mathrm{kg \cdot m^2}$
n_N	Rated speed	2000 rpm
T_{LN}	Rated torque	10 $\mathrm{N \cdot m}$
$I_{\max}$	Maximum current	10 A
U_{dc}	DC bus voltage	300 V

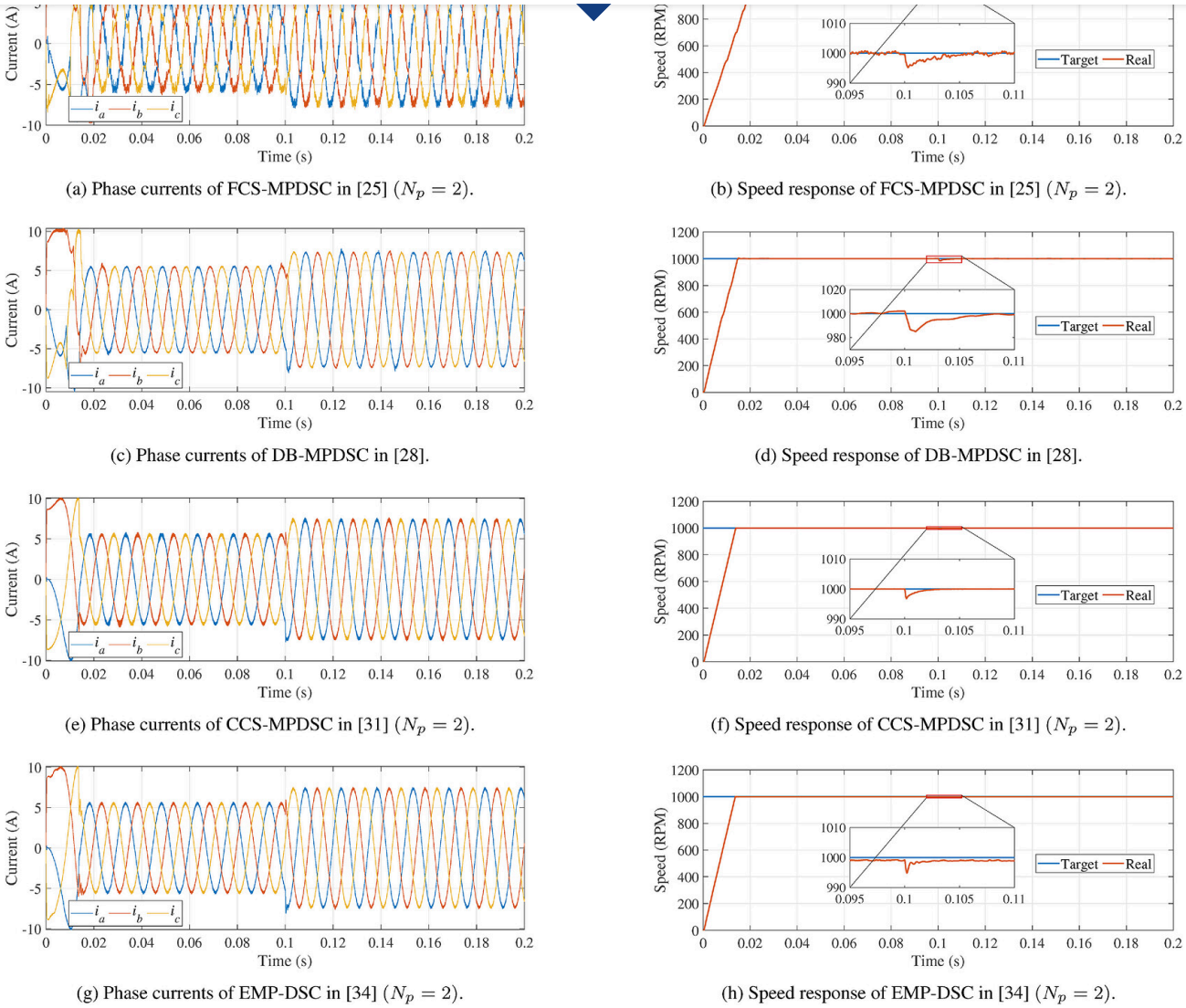
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T_s	Electrical sampling period	50 μs
T_{sv}	Mechanical sampling period	500 μs

TABLE 9. Online computational burden and THD comparison.

Methods	Time (μs)	THD (%)
FCS-MPDSC in [25]	18.44	9.28
DB-MPDSC in [28]	12.59	2.13
CCS-MPDSC in [31]	45.76	1.31
EMP-DSC in [34]	9.34	1.73

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(a) Phase currents of FCS-MPDSC in [25] ($N_p = 2$).

(b) Speed response of FCS-MPDSC in [25] ($N_p = 2$).

(c) Phase currents of DB-MPDSC in [28].

(d) Speed response of DB-MPDSC in [28].

(e) Phase currents of CCS-MPDSC in [31] ($N_p = 2$).

(f) Speed response of CCS-MPDSC in [31] ($N_p = 2$).

(g) Phase currents of EMP-DSC in [34] ($N_p = 2$).

(h) Speed response of EMP-DSC in [34] ($N_p = 2$).

FIGURE 13

Open in figure viewer | PowerPoint

Simulation results.

Additionally, to compare the online computational burden of proposed schemes, real-time hardware-in-the-loop (HIL) experiments of the core algorithm are conducted on an STM32G4 series micro-controller, with results shown in Table 9. It should be noted that the time-consumption values listed in Table 9 include essential processes such as sensor self-diagnostics, thus can only relatively reflecting the computational burden difference among schemes.

Basically, all schemes achieve effective speed tracking. Schemes in [31, 34] employ SVPWM, resulting in significantly reduced current THDs. [25, 28] utilize single-vector and dual-vector synthesis respectively, leading to higher current THDs and larger speed ripples. The FCS-MPDSC scheme in [25] exhibits fast dynamic response but with the highest current THD. For the FCS method in [25], when the speed error is sufficiently small, the current cost term dominates,

tracking errors, with its current performance also being slightly inferior to that of CCS-MPDSC in [31]. All schemes incorporate observers for load torque estimation to ensure the system robustness, resulting in a swift speed tracking response after the torque disturbance.

For the computational burden compared in Table 9, DB-MPDSC in [28] imposes the lowest burden. The FCS-MPDSC scheme in [25] incurs slightly higher costs due to the need to traverse 16 voltage vector combinations from the finite control set. CCS-MPDSC in [31] bears the highest burden from solving of high-dimensional optimization problems online. In contrast, EMP-DSC in [34] achieves the minimal online burden through offline optimization. However, it requires higher memory and brings greater offline computational burden.

8.2 Overall Analysis

A comprehensive comparison across four MPDSC categories is presented in Table 10. In essence, FCS-MPDSC is enumerative optimization, DB-MPDSC is analytic inversion, CCS-MPDSC is online constrained optimization and EMP-DSC is pre-solved optimization.

TABLE 10. Comprehensive comparison across four MPDSC categories.

Category	Computational burden	Real-time capability	Constraint handling	Hardware needs	Key advantages	Key drawbacks
FCS-MPDSC	Low to medium	Good	Strong	Low to moderate	Flexible design	Ripples
DB-MPDSC	Low to medium	Good	Weak	Moderate	No weighting factors	Constraint handling
CCS-MPDSC	High	Moderate	Strong	High	High performance	Online burden
EMP-DSC	Very low online; high offline	Best online	Strong	Moderate to high	High consistency	Offline complexity

Regarding typical application scenarios, FCS-MPDSC is well-suited to medium- or high-power motor drives, such as EV traction inverters, wind generators and industrial drives, where switching loss management and multi-objective control are important and embedded computing resources are limited. It is less ideal in situations requiring low current ripple, such as precision servo positioning.

DB-MPDSC directly computes voltage reference values based on a discrete-time model, thereby avoiding online iterative optimization. It is well-suited to applications requiring high dynamic response with modest demands on multi-objective optimization and constraint handling, such

significantly limited in complex scenarios involving strong constraint coupling or coordinated multi-objective control.

CCS-MPDSC is well-suited to scenarios requiring superior current quality and sufficient computing power, for example, precision servo drives and high-end industrial robotics. It performs outstandingly when constraints dynamically change with operating conditions and necessitate explicit enforcement. Nonetheless, the online QP burden constrains the prediction horizon on budget micro-controllers, making it suboptimal for cost-sensitive and high-volume embedded implements.

EMP-DSC is suitable for systems with extremely tight online timing budgets and strict constraint compliance, such as automotive electronic control units (ECUs) and aerospace drives. The separation of offline computation and online execution makes it attractive for mass production where the plant model is fixed and the controller is repeatedly deployed. This approach is less suitable in scenarios with limited memory and computational resources.

9 Future Trends

9.1 Refinement of Stability Theory

Lyapunov theory is an important tool for stability analysis. Accordingly, some studies construct cost functions via Lyapunov-based methods so that their forms satisfy stability requirements [182, 183]. For direct torque control, [184] employs a Lyapunov approach to tune the weighting factors in the cost function, and [79, 129] extend this idea to MPDSC. [185] proposes an improved Lyapunov-function constraint that allows non-monotonic convergence to a terminal set. In [186], a stability strategy for FCS-MPC is developed by integrating a Lyapunov controller with virtual voltage vectors, enhancing disturbance rejection and system stability.

To address the stability of FCS-MPC in the presence of time-varying disturbances, [187] proposes an analysis and design framework that integrates a disturbance observer with a sequence of robust positively invariant sets. By designing time-varying, contractive constraint sets to handle estimation and quantization errors, recursive feasibility and practical asymptotic stability of the closed-loop system are ensured. Based on regional input-to-state stability theory, [188] derives sufficient conditions for guaranteeing regional stability and establishes methods to estimate the system's domain of attraction and the ultimately bounded set.

Nevertheless, theoretical proofs for the overall framework remain insufficient. Related work has only been partially investigated as recently as 2025, which makes it difficult to provide stability-oriented guidance for the overall design.

9.2 Multi-Step Prediction

For FCS-MPDSC, CCS-MPDSC and EMP-DSC, multi-step prediction constitutes an important direction for improving the overall performance of MPDSC. Further investigations are still

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In existing studies, reducing the number of switching combinations under multi-step prediction is often achieved at the cost of control performance. For common studies, the switching combinations of two consecutive steps are constrained to differ by at most one leg, thereby halving the number of candidates [53, 54]. Some studies adopt less reasonable schemes, such as forcing the second-step switching vector to be the same as that in the first-step [27]. In [189], the current response for the next step is predicted in order to select the optimal vector over two control periods. In [190], all vectors are enumerated in the first step, so that the optimal and suboptimal vectors are selected, and further used as the only candidates for subsequent step. Under these schemes, current harmonics can be suppressed, but the dynamic performance is compromised. Therefore, it is necessary to develop an adaptive candidate set reduction strategy accounting for both dynamic and steady-state performance.

Moreover, in multi-step prediction, the second-step optimal control action is often overlooked. In [66], the selected second-step optimal voltage vector is encouraged in the cost function when the system is under steady-state conditions. However, the coefficient of the encouraged term can only be set based on experience, and the number of switching combinations to be evaluated is not reduced. How to reduce the computational burden while fully utilizing the information contained in the second-step optimal vector remains an open problem.

For the high-dimensional optimization problem in multi-step prediction, faster solver algorithms or approximate solution strategies can be employed to reduce the computational burden. However, how to properly balance solver capability and computational burden still deserves further study.

9.3 Dimension Reduction for EMP-DSC

EMP-DSC transforms the online optimization into a lookup table of a piecewise affine mapping. Compared with other MPDSC categories, the online computational burden is substantially reduced. Nevertheless, EMP-DSC still suffers from the problem of dimensionality and parameter dependence. If all relevant parameters are fully taken into account, the system state vector can reach at least nine dimensions [112, 113].

A cascaded architecture is proposed in [114], with the speed loop and the current loop designed separately, reaching a relatively low state dimension for each individual loop. However, this architecture essentially departs from the concept of direct speed control. [115, 116] neglected the current coupling terms, with a limited reduction in dimension, and at the cost of accuracy.

Reducing the state vector dimension can not only significantly decrease the difficulty of solving the optimization problem, but also reduce the memory usage for storing the explicit solution, thereby fully demonstrating advantages of EMP-DSC. Therefore, developing a method that can both realize a high-bandwidth direct speed control and reduce the dimension is expected to become a focused research topic.

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With rapid advances in deep learning, reinforcement learning, and big-data analysis, the integration of data-driven methods with MPC is becoming a major trend in intelligent electric-drive systems.

As previously discussed, data-driven methods have been employed for weight factor tuning and other tasks in MPC for PMSMs [132, 134, 135]. In recent years, MFPC has emerged as a prominent research direction in power electronics. [191, 192] propose control strategies for dual active bridge (DAB) converters from traditional FCS and CCS perspectives. Beyond these efforts, advanced data-driven methods have been increasingly integrated into current control schemes [193-195].

Implementing data-driven methods in current loop requires enhanced circuit topologies and superior processor performance, resulting in a limited number of relevant studies. But it should be noted that integrating data-driven methods into direct speed control is highly promising. Furthermore, due to the independence from precise physical models for MFPC, the optimization complexity does not increase correspondingly with the model order. As a result, it theoretically affords greater development potential for the transition from MFPCC to MFPDSC than from MPCC to MPDSC.

To address the black-box characteristic and potential safety risks of data-driven methods, it is necessary to incorporate interpretability analysis, robustness evaluation, and safety-constraint design. In this way, a model-based data-driven collaborative MPDSC scheme can be established, which improves performance while ensuring that the control policy remains verifiable, diagnosable and maintainable.

9.5 Application in Other Machine Types

Although this paper primarily focuses on the control of PMSMs, recent years have witnessed preliminary explorations of MPDSC in other machine types, including synchronous reluctance motors (SRMs) [127], double fed induction generators (DFIGs) [196, 197], induction motors (IMs) [198-200], and magnetic levitation transportation long stator linear motors (MTLSLMs) [201]. [198] proposes a direct speed predictive control strategy for IMs, achieving sensorless direct speed regulation by incorporating an EKF observer. Subsequently, a decision method based on the Euclidean norm is introduced to eliminate the complexity of weighting factor design [199]. [201] addresses the time-varying parameters of MTLSLMs by proposing a MFPDSC strategy combined with a moving horizon estimation (MHE) observer for disturbance compensation, simultaneously achieving neutral-point potential balancing for three-level inverters.

However, compared with PMSMs, MPDSC research with application to other machine types remains relatively limited. IMs face challenges regarding strong nonlinearity and flux observation, MTLSLMs exhibit significant parameter time-variance and DFIGs must coordinate grid synchronization with rotor dynamics. Consequently, MPDSC in these scenarios needs further research. Furthermore, for example, PMSMs and IMs coexist in the multi-motor

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integration and control performance.

10 Conclusion

Precision servos play a significant role in the advancement of modern industry. This paper presents the first systematic review of MPDSC methods for PMSMs, which has gained increasing research attention in recent years.

Following a retrospective of several pioneering studies, existing MPDSC approaches are categorized into four main types. Their respective optimization strategies are discussed. Subsequently, solutions to weighting-factor and robustness problem are summarized. Experimental comparisons are designed for classical schemes of each method, accompanied by theoretical analysis and discussions on application scenarios. Overall, MPDSC breaks away from the traditional cascaded control structure, supports multi-objective optimization and substantially enhances system bandwidth. Finally, existing challenges within MPDSC are analysed, based on which several valuable directions for future research are proposed.

We sincerely hope the paper could contribute to the further refinement of this theory and promote its recognition and application in engineering practice.

Author Contributions

Hao Fan: conceptualization, investigation, methodology, writing – original draft, writing – review and editing. **Tianzhe Yang**: investigation, validation, visualization, writing – review and editing. **Baoyang Liu**: data curation, visualization. **Chaochen Gu**: project administration, supervision. **Xinping Guan**: supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Supporting Information